

A provenance-sealed DESI DR1 anomaly-score candidate catalogue: selection, validation, and a null known-object recovery benchmark

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No discovery is claimed in this paper: we publish a machine-flagged candidate catalogue of spectroscopically unusual objects in Data Release 1 of the Dark Energy Spectroscopic Instrument (DESI), together with the complete provenance chain needed to reproduce the catalogue’s selection (score, provenance gate, threshold), and a known-object recovery benchmark that clears none of its pre-declared bars. One downstream, non-selection analysis — the descriptive taxonomy of Sec. VI — is fully specified and reproducible from the released files (spherical sky embedding, stated hyperparameters and seed), carrying its own separate provenance record distinct from the sealed selection chain (Sec. VI). An archived convolutional autoencoder, bound by SHA-256 to a sealed run contract, assigns a reconstruction-error anomaly score S to every science-target spectrum in the public `iron` release; 28,425,963 raw rows deduplicate to 27,547,223 unique `TARGETIDS`, of which 21,793,550 pass the science-target provenance gate. We show that a score cut alone does not select astrophysical objects: above $S = 3$, more than 99.5% of raw fibres in every score bin are sky or non-science fibres, and that fraction rises with score up to $S = 10$, above which the single remaining bin (337 fibres, one science target) is too sparse to constrain the trend. Applying the provenance gate (`OBJTYPE = TGT`, `COADD_FIBERSTATUS = 0`, `TARGETID > 0`) before any score cut, and a pre-declared threshold rule, yields a released catalogue of 1,244 objects — the top 5.7×10^{-5} of the science-target population. We validate the catalogue against a published contract of sixteen machine-checkable requirements (eleven pass outright; two are reported quantities rather than thresholds; one returns a weak positive signal; two fail by design, to force the release to state its own defects), and report three results that constrain its interpretation. First, the score is essentially uncorrelated with exposure quality and source brightness (Spearman $|\rho_s| \leq 0.10$ against five independent measures), so the tail is not a faintness or short-exposure artefact. Second, the score is not merely correlated with but almost entirely *explained by* the blue spectrograph arm’s own reconstruction residual: it is the dominant term of the score by construction (94.9% of released objects have it as their worst-fit camera; the full three-camera regression explains $R^2 = 0.78$ of the score’s variance and collapses to $R^2 = 0.01$ when it is dropped, while it alone, univariately, explains $R^2 = 0.20$), the residual shows no dependence on blue-arm signal-to-noise ($\rho_s = -0.01$), and its wavelength distribution concentrates far more tightly in the observed frame than the rest frame for spectroscopically confirmed objects — evidence favouring a fixed-wavelength calibration origin over a genuine spectral feature, though we do not close the question here. Third, the catalogue’s taxonomy of 23 clusters and 9 families, built over the 675 objects with no SIMBAD or NED counterpart, is structured but not separated in the model’s own latent space and must be read as a stratification of the candidate list rather than as physical classes. A known-object recovery benchmark against five reference classes recovers no class at its pre-declared bar; with reference samples of 27–5,285 objects the benchmark has little power (95% upper limits up to 12.5% recovery for the rarest class) and excludes only a strong recovery signal, so this work is reported as a validated data release and not as a discovery. We name 30 follow-up targets in five rules-based tiers and run the cheapest of the named tests here: of four anomaly-selected DESI-pipeline $z \geq 4$ quasar candidates with no catalogued counterpart, the released Legacy Survey photometry already refutes one on public data alone, and we can neither confirm nor refute the remaining three, because the published schema omits flux inverse variances and because one candidate’s own photometry sits inside a disclosed photometric-join defect — among the release defects we record openly rather than hide. We close with the open tests, led by the absent selection function, that any population-level use of this catalogue must clear first.

I. INTRODUCTION

Wide-field spectroscopic surveys produce far more spectra than can be inspected. Unsupervised outlier detection is the standard response [10], and it works: a model trained to reconstruct typical spectra flags the ones it cannot. The approach has been carried into DESI

itself, both on the Bright Galaxy Survey [11] and with variational autoencoders on galaxy spectra [12].

What such a search usually does not publish is the part that decides whether its output means anything — the exact model, the exact code, the exact parent catalogue, the selection rule, the reason the rule was chosen, and the tests the resulting list did *not* pass. Without those, a candidate list is unfalsifiable: a reader cannot tell a genuine spectral outlier from a fibre placed on blank sky, and cannot reproduce either.

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This paper publishes that part alongside the catalogue. Our contributions are:

1. **A sealed provenance chain** (Sec. IIB): the SHA-256 of the parent zcatalog, of the archived model, of the inference code, and of every intermediate product, bound under one run contract and re-verified end to end at publication time.
2. **A demonstration that the provenance gate, not the score cut, is what makes such a sample astrophysical** (Sec. IIIB). Above $S = 3$ the raw fibre population is more than 99.5% sky or non-science in every score bin, and the contamination *worsens* with score up to $S = 10$, above which the single remaining bin is too sparse (337 fibres, one science target) to constrain the trend. This is a property of the substrate, not of our model, and it applies to any reconstruction-error search run on public coadds.
3. **A published validation contract** of sixteen machine-checkable requirements (Sec. V), with its checker and raw output released, including four release defects the release carries openly rather than hides.
4. **Three measurements of what the score responds to** (Secs. VB–VC, VIC): it is not an exposure-quality proxy; it is almost entirely explained by the blue-arm reconstruction residual (full-model $R^2 = 0.78$, collapsing to 0.01 without it; the blue arm alone, univariately, explains $R^2 = 0.20$), whose own wavelength distribution favours an instrumental over an astrophysical origin without closing the question; and the taxonomy it supports is descriptive rather than physical.
5. **A negative, low-power recovery benchmark** (Sec. VD): against five published reference classes, no class is recovered at the pre-declared bar, and the benchmark’s small reference samples give it little power to have found one (95% upper limits up to 12.5% recovery for the rarest class).
6. **A named, rules-based follow-up set** (Sec. VII), with the cheapest test in the set — a Lyman-break consistency check on public imaging — executed here rather than deferred.

The framing follows from the benchmark. A catalogue whose pre-declared success condition is unmet is a data release with a validated selection, not a discovery paper, and we report it as one throughout. Section II describes the data and the provenance chain, Sec. III the model and the selection rule, Sec. IV the catalogue itself, Sec. V the validation contract and what it establishes, Sec. VI the descriptive taxonomy, Sec. VII the named follow-up set and the open tests, Sec. VIII what this catalogue does and does not bear on cosmologically, Sec. IX the reproduction path, and Sec. X what this catalogue is not.

II. DATA

A. DESI DR1

We use Data Release 1 of DESI [13, 14]: the public `iron` spectroscopic reduction, its redshift catalogue `zall-pix-iron.fits`, and the per-(survey, program, healpix) coadd spectra, streamed and scored in place rather than bulk-downloaded. Redshifts, spectral classifications and quality flags are the pipeline’s own, fit by Redrock [9] and documented in [15, 16].

The scan covered a 36,634-group locator inventory, receipt-verified group by group, and processed 28,425,963 raw rows. Deduplication by last `TARGETID` occurrence in lexical shard-then-row order removed 878,740 duplicate rows and left 27,547,223 unique `TARGETID`s. Of those, 21,793,550 satisfy the science-target provenance gate defined in Sec. IIIB; that number, not the full 27.5 million, is the parent population of this catalogue, and every selected fraction quoted below uses it. The deduplication rule itself has no physical content: which of a repeated `TARGETID`’s several DESI (survey, program) observations survives depends on shard file names and row order, not on which exposure is better. We have not quantified how many of the 1,244 released objects sit near enough to threshold that their alternate duplicate would score below it, and record this as an open, unquantified caveat on near-threshold catalogue membership rather than assert it is small.

B. Provenance chain

Every stage binds the exact SHA-256 of its input and output under a single sealed run contract (generation `clean-rerun-6699d09ff886`). The thirteen hash-bound artifacts are listed in Table I: the parent zcatalog, the run contract itself, the archived model and inference code, and the nine released data products.

All eight links of the manifest chain — sample $\rightarrow$ enrichment $\rightarrow$ cross-match $\rightarrow$ taxonomy, each stage naming its input hash and its output hash — are verified in the validation contract (check V3, Table VI), and all nine released artifacts re-hash to the values in the landing receipt (check V2).

III. METHOD

A. The archived model and its calibration

Spectra are scored by an archived autoencoder frozen for this run. The input is a per-spectrum median-absolute-flux-normalised 496-bin resampling; the encoder compresses it to 128 latent dimensions, and the score is

TABLE I. The hash-bound provenance chain. Nine released data products are re-hashed at publication time and matched against the landing receipt (check V2); the run contract, archived model and parent zcatalog hashes are recorded and chain-bound (check V3); the inference-code hash is recorded in the run contract.

Artifact	SHA-256 (first 16 hex)
DESI DR1 zcatalog <code>zall-pix-iron.fits</code>	2d95ad99361039b5...
sealed run contract (<code>clean-rerun-6699d09ff886</code>)	6699d09ff886f74d...
archived model <code>best_model.47k.pt</code>	f5266ba48f476bca...
inference code	3e7efb243fa5cc4e...
<code>flagship_crossmatch_v2_matched.parquet</code>	4a718d1a7ad253d2...
<code>flagship_crossmatch_v2_unmatched.parquet</code>	c0a6b57bd672f81b...
<code>flagship_sample.v2.parquet</code>	d6d43dfa04d6a8b2...
<code>flagship_sample.v2.enriched.parquet</code>	c3b176ff2d355a42...
<code>flagship_taxonomy_v2.json</code>	1420388b59f37278...
<code>flagship_wise_v2.parquet</code>	1b42125d5e62830c...
<code>science_target_summary.json</code>	439f886bd927625d...
<code>sky_fraction_by_score.json</code>	b2741b18dd543157...
<code>threshold.choice.json</code>	e4244bbab5779c87...

a standardised reconstruction-error statistic,

$$S = \frac{\overline{\text{MSE}} - \mu_{\text{MSE}}}{\sigma_{\text{MSE}}}, \quad (1)$$

with $\overline{\text{MSE}}$ the per-spectrum mean squared reconstruction error and $(\mu_{\text{MSE}}, \sigma_{\text{MSE}}) = (0.8771, 1.3605)$ sealed before the scan. A high S therefore means a spectrum unlike the shapes the model learned to reproduce; it is a distance in the model’s own reconstruction metric and carries no physical normalisation.

The model itself, `best_model.47k.pt`, predates the clean-rerun contract this release is bound under and is treated here strictly as a frozen, hash-sealed artifact (Sec. IIB): its weights are re-verified byte for byte, but its training corpus (parent survey, target classes, selection, any quality cuts, and any overlap with the 21.8 million science targets scanned here), its wavelength-to-bin mapping across the three cameras, and its architecture beyond “convolutional autoencoder, $496 \rightarrow 128$ ” are not documented in any artifact this release controls. Two consequences follow directly from Sec. VC: if the training corpus is drawn from the same imaging/spectroscopic population scored here, objects inside it would carry an artificially low reconstruction error, a selection effect operating on the exact quantity that defines the catalogue; and the per-camera bin count is a candidate contributor, alongside the calibration explanation of Sec. VC, to why the reconstruction loss is dominated by the b camera. Neither can be assessed without the missing documentation, which we did not reconstruct or guess at for this release; closing it is a named prerequisite for any use of this catalogue beyond the data-release-with-caveats framing adopted here.

The calibration constants were fit on a two-stage probability-proportional-to-size cluster sample of 200 groups and 20,000 held-out spectra, with a further 20,000

reserved for a stability check. The check requires $|\mu_{\text{MSE}}^{\text{val}} - \mu_{\text{MSE}}| \leq 5 \sigma_{\text{MSE}}/\sqrt{n}$; the observed deviation was 0.00543 against a bound of 0.0481, and it passed. The constants and the model weights were then frozen and hash-bound before any spectrum of the survey was scored, so the score of a given spectrum is fixed by the contract rather than by the sample it is scored with.

Across the full scan the score distribution is sharply peaked and long-tailed: over all fibres the median is -0.56 , the 99th percentile is 3.08, and the maximum observed is 24.42. Over science targets alone the distribution is tighter — median -0.60 , 99th percentile 0.96, maximum 11.33. The released threshold of $S > 3$ therefore sits far above the science-target 99th percentile, and the released catalogue is the top 5.7×10^{-5} of the science-target population.

B. The science-target provenance gate

A row enters the candidate pool only if `OBJTYPE = TGT` and `COADD_FIBERSTATUS = 0` and `TARGETID > 0`, the last asserted rather than merely filtered. This gate is applied *before* any score cut and is enforced by a committed, re-runnable checker, [check_sample_provenance.py](#) (full path in the reproducibility manifest).

The gate is not a formality, and the quantitative reason is the single most transferable result in this paper. Figure 1 and Table II give the fraction of raw fibres that are sky or non-science as a function of score: it exceeds 99.5% in every bin above $S = 3$ and *increases* monotonically with score, from 99.53% in $[3, 4)$ to 99.97% in $[8, 10)$; the sparsely populated top bin (337 fibres, one science target) sits at 99.70%. A naive top-score selection from this scan returns instrument artefacts. The effect is a property of the public coadd substrate — sky fibres

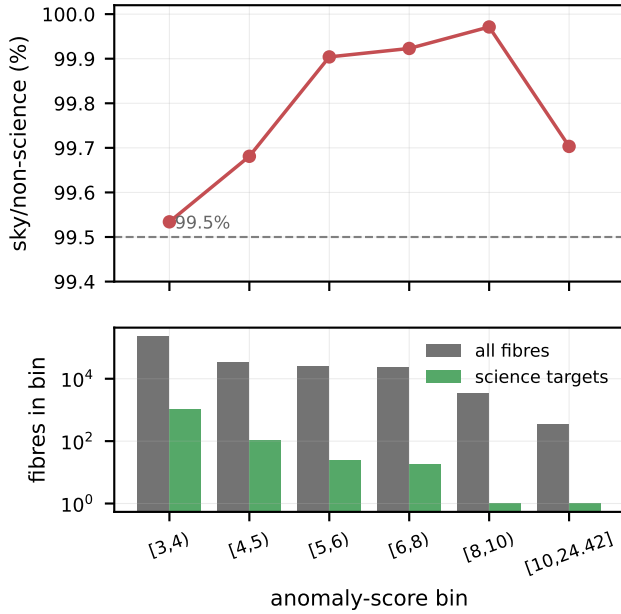


FIG. 1. *Top*: sky-or-non-science fibre fraction per score bin, before the provenance gate; note the suppressed zero. The fraction exceeds 99.5% in every bin and rises monotonically to 99.97% in [8, 10). *Bottom*: the bin populations behind it. Redrawn from the released `sky_fraction_by_score.json`.

TABLE II. Per-bin fibre census before the provenance gate. “Sky/non-science” is NOT(OBJTYPE = TGT AND COADD_FIBERSTATUS = 0), with TARGETID ≤ 0 counted as sky/non-science.

Score bin	All fibres	Science targets	Sky/non-science
[3, 4)	234,365	1,092	99.53%
[4, 5)	33,865	108	99.68%
[5, 6)	25,008	24	99.90%
[6, 8)	23,370	18	99.92%
[8, 10)	3,473	1	99.97%
[10, 24.42]	337	1	99.70%

are reconstructed worst precisely because they contain no source — so any reconstruction-error search on DESI coadds that omits a provenance gate will report the same population, whatever its model.

C. The threshold rule

The release threshold follows a rule fixed before the science-only counts were inspected: from the grid {3, 4, 5, 6, 8, 10}, take the largest threshold whose science-only count is ≥ 300; if that count exceeds 1,500, step to the next grid point unless doing so drops the count below 300. The rule selects $S > 3$ with $n = 1,244$, reproduced independently from the counts rather than read from the

TABLE III. Threshold ladder: all fibres versus science targets.

$S >$	All fibres	Science targets	Science fraction
3	320,418	1,244	0.39%
4	86,053	152	0.18%
5	52,188	44	0.08%
6	27,180	20	0.07%
8	3,810	2	0.05%
10	337	1	0.30%

stored choice in check V4. Table III gives the raw and science-target counts on the ladder; Fig. 2 plots both survival curves alongside the released sample’s score distribution. The two curves diverge by more than two orders of magnitude, which is the same statement as Fig. 1 read cumulatively.

IV. THE CATALOGUE

A. Contents

The released catalogue contains 1,244 objects. Table IV lists the published columns by group: identification and sky position, the anomaly score with its per-camera residual decomposition, the 128 latent activations, the DESI pipeline redshift solution, exposure-quality diagnostics, Legacy Survey photometry [23], the AllWISE [8] and SIMBAD/NED cross-match results, the taxonomy labels of Sec. VI, and a group of derived colours, boolean flags, and join tags added at assembly time: `mean_fiber_ra/mean_fiber_dec` (DESI’s own per-object mean fibre position across the coadded exposures); `gr_color`, `rz_color`, `w1w2_color` (AB-magnitude colours computed from the Legacy Survey and AllWISE fluxes already listed above); `is_point_source` (`morphtype==‘PSF’`); `is_star_candidate` (`parallax > 0.5 mas`, or `gr_color < 0.5` and `rz_color < 0.3` jointly, a stellar-locus colour cut); `crossmatch` (the matched/unmatched tag assigned by the SIMBAD/NED join); and `snr_med` (the per-object median of the three `median_coadd_snr_{b,r,z}` columns, used by Sec. VC’s diagnostics). All nine are deterministic functions of already-released columns, defined in the committed `enhanced_18M_inference.py` (colours, flags) and `assemble_flagship_evidence.py` (`crossmatch`, `snr_med`).

Scores range from 3.0002 to 11.3316, with a median of 3.28 and a 90th percentile of 4.08: the catalogue is dominated by objects just above threshold, as any tail-selected sample is.

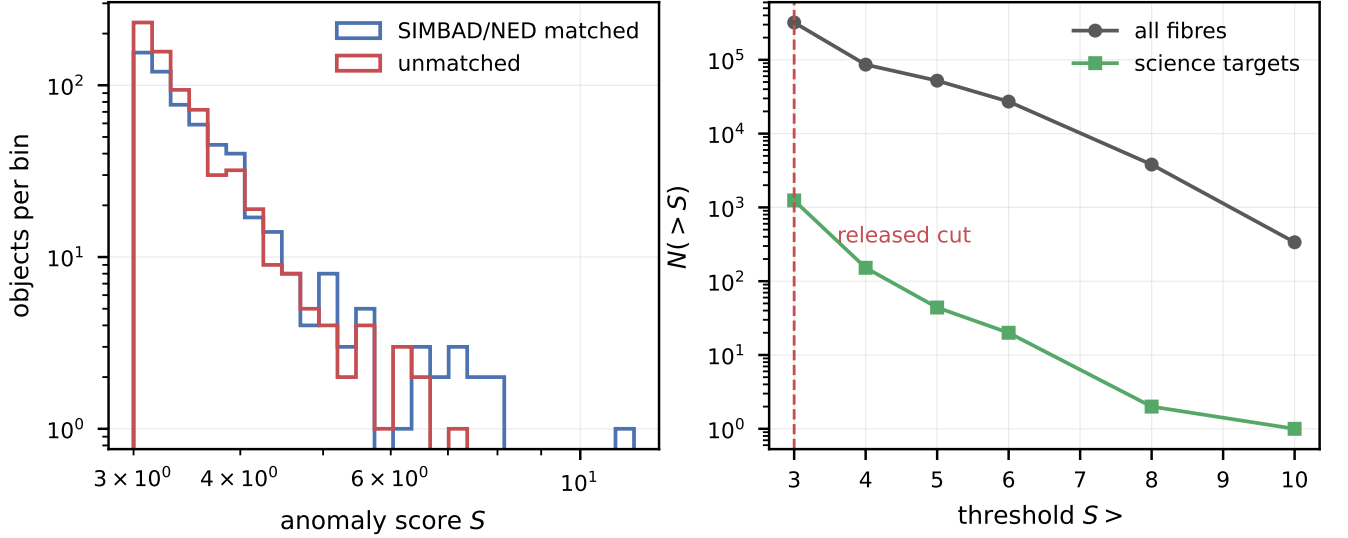


FIG. 2. *Left*: anomaly-score distribution of the 1,244 released objects, split by whether a SIMBAD or NED counterpart exists. *Right*: survival counts $N(> S)$ on the pre-declared threshold grid for all fibres and for science targets; the released cut is marked. The gap between the two curves is the sky-fibre population removed by the provenance gate.

TABLE IV. Released column schema, by group: all 192 released columns accounted for. `classification` and `discovery_potential` are constant placeholders across all 1,244 rows, listed rather than dropped so the schema stays honest about what is not yet populated.

Group	N_{col}	Columns
identification	6	<code>targetid</code> , <code>target_ra</code> , <code>target_dec</code> , <code>survey</code> , <code>program</code> , <code>healpix</code>
anomaly	8	<code>anomaly_score</code> , <code>mean_mse</code> , <code>rB</code> , <code>rR</code> , <code>rZ</code> , <code>worst_band</code> , <code>peak_residual_wavelength</code> , <code>residual_kurtosis</code>
latent	128	<code>latent_000</code> – <code>latent_127</code>
DESI pipeline	4	<code>z</code> , <code>zwarn</code> , <code>spectype</code> , <code>deltachi2</code>
exposure	9	<code>coadd_numexp</code> , <code>coadd_exptime</code> , <code>median_coadd_snr_b</code> , <code>median_coadd_snr_r</code> , <code>median_coadd_snr_z</code> , <code>tsnr2_qso</code> , <code>tsnr2_elg</code> , <code>tsnr2_lrg</code> , <code>tsnr2_bgs</code>
Legacy Survey photometry	11	<code>flux_g</code> , <code>flux_r</code> , <code>flux_z</code> , <code>flux_w1</code> , <code>flux_w2</code> , <code>morphtype</code> , <code>sersic</code> , <code>shape_r</code> , <code>ebv</code> , <code>parallax</code> , <code>gaia_phot_g_mean_mag</code>
AllWISE cross-match	5	<code>w1</code> , <code>w2</code> , <code>w1_w2</code> , <code>match_separation_arcsec</code> , <code>match_flag</code>
SIMBAD/NED cross-match	6	<code>simbad_found</code> , <code>simbad_main_id</code> , <code>simbad_otype</code> , <code>ned_found</code> , <code>ned_name</code> , <code>ned_type</code>
taxonomy	4	<code>cluster_id</code> , <code>family_id</code> , <code>family_descriptor</code> , <code>is_core_member</code>
derived colours, flags & join tags	9	<code>mean_fiber_ra</code> , <code>mean_fiber_dec</code> , <code>gr_color</code> , <code>rz_color</code> , <code>w1w2_color</code> , <code>is_point_source</code> , <code>is_star_candidate</code> , <code>crossmatch</code> , <code>snr_med</code>
unpopulated placeholder	2	<code>classification</code> , <code>discovery_potential</code>

B. Redshift reliability

Of the 1,244, 502 (40.4%) carry a reliable pipeline redshift (`ZWARN` = 0); the remaining 742 carry `ZWARN` = 4 (`SMALL_DELTA_CHI2`) and their redshifts must not be used as measurements [16]. Every redshift-dependent statement in this paper is restricted to the `ZWARN` = 0 subset, which has median $z = 1.14$ and splits 424 `GALAXY` / 73 `QSO` / 5 `STAR`; 70 objects sit at $z > 2$, 36 at $z \geq 4$ and 16 at $z \geq 6$, with a maximum of $z = 6.75$. Pipeline classifications over the whole sample are 1,153 `GALAXY`, 86 `QSO`

and 5 `STAR`. Figure 3 shows the reliable-redshift distribution and its relation to the score; there is no visible trend of score with redshift ($\rho_s = 0.009$, $p = 0.85$, $n = 502$), and the high-redshift objects are not the high-score ones.

C. Photometry and morphology

Legacy Survey photometry is carried for every object. 441 are typed `PSF` (point-like), 316 `REX`, 61 `EXP`, 14 `SER`, and 6 `DEV`, with 406 rows carrying an empty `MORPHTYPE`

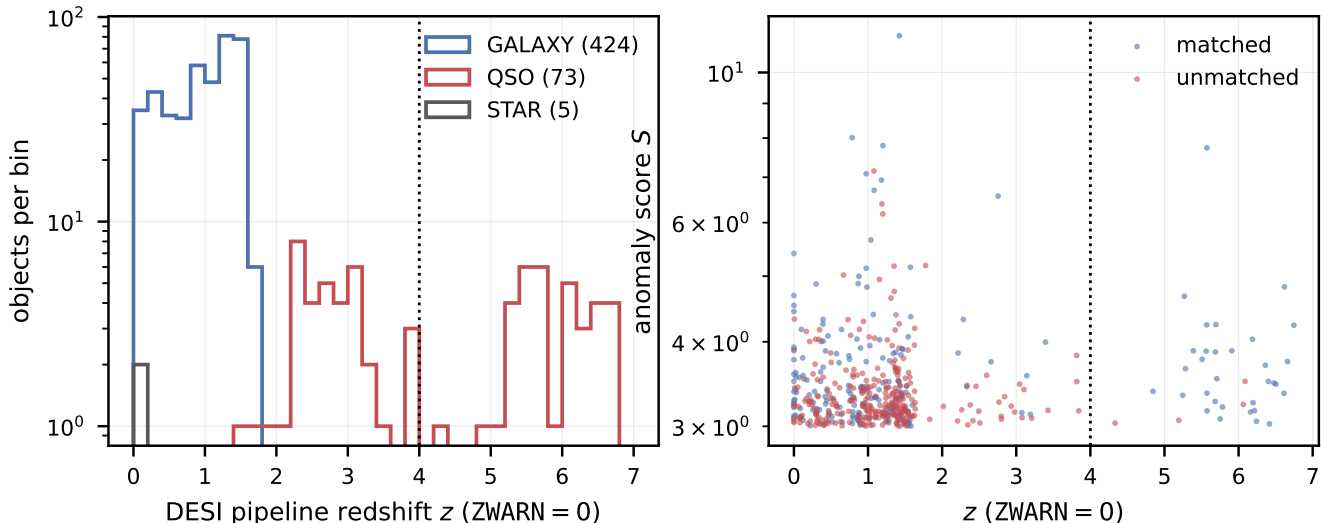


FIG. 3. *Left*: DESI pipeline redshifts of the 502 objects with $ZWARN = 0$, by pipeline classification; the $z = 4$ line marks the FT-A tier boundary. *Right*: score against redshift for the same subset. High score and high redshift are not the same population.

— a gap in the photometric join that the deposit must close. The six values sum to the full released 1,244. Only 6 objects have a Gaia DR3 counterpart with a measured G magnitude [24] and only 5 a non-zero parallax, so the sample is, with a handful of exceptions, fainter than Gaia’s limit; none of the 675 unmatched objects has a Gaia counterpart at all. The median coadd exposure time is 1,129 s. AllWISE counterparts exist for 74 objects, with median $W1 - W2 = 0.51$.

The released schema does *not* carry flux inverse variances. That omission is consequential — it prevents a reader from forming a detection significance for any photometric statement, including the Lyman-break test of Sec. VII B — and it is recorded here as a release defect to be repaired in the public deposit.

D. External counterparts

SIMBAD [17] and NED [6] cross-matching at $3''$ (queried through astroquery [37], as is VizieR [7] for AllWISE below) identifies counterparts for 569 objects (45.7%): 562 via NED, 38 via SIMBAD, 31 via both. That leaves 675 (54.3%) unmatched. Table V breaks the matches down by counterpart type. They are overwhelmingly ordinary: 455 of the NED matches are typed **G** (galaxy) and 52 **IrS** (infrared source), with 31 **QSO**; the SIMBAD side, which matches far fewer objects, is relatively richer in quasars (19 of 38).

Unmatched does not mean new. The unmatched set includes objects simply outside other surveys’ footprints or below their detection limits, and the Gaia statistic above shows the population is faint. The extreme tail sharpens this: the six highest-scoring objects in the whole sample

TABLE V. Counterpart types of the 569 matched objects, listed per service (columns are not disjoint: 31 objects match both, so NED’s 562 and SIMBAD’s 38 sum to 600). All 8 distinct NED types and all 6 distinct SIMBAD types are shown; nothing is truncated. NED types dominate by count; SIMBAD matches fewer objects but a larger quasar fraction.

NED type	N	SIMBAD type	N
G	455	QSO	19
IrS	52	G	12
QSO	31	*	4
*	14	PN	1
VisS	4	rG	1
UvS	3	WD?	1
RadioS	2		
XrayS	1		

($S = 11.33, 8.01, 7.80, 7.74, 7.57, 7.33$) are all *matched*, i.e. already catalogued, and matched objects score higher than unmatched ones at the population level too (median 3.337 vs. 3.237, Mann–Whitney $p = 4.5 \times 10^{-5}$). We do not read this as a signal about the method, because it is confounded with brightness: SIMBAD/NED-matched objects are also significantly brighter and better measured than unmatched ones (median r -band flux 0.289 vs. 0.184 nmgy, $p = 8 \times 10^{-4}$; median coadd r -band signal-to-noise 0.079 vs. 0.046, $p = 1 \times 10^{-6}$) — exactly what makes an object catalogued elsewhere in the first place. A brightness preference in the score would produce this same matched/unmatched difference with no implication for whether the score finds anything astrophysically unusual, and Sec. V B already shows the

score is not a brightness proxy across the released sample, not within this split specifically. The decisive evidence remains Sec. VD: the score preferentially selects catalogued objects, but recovers no specific class of genuinely unusual object at any enrichment. The reminder this leaves stands: the unmatched subset is not the interesting tail by construction.

Figure 4 shows the sky distribution of both subsets. The catalogue traces the DR1 footprint and its tiling pattern; neither subset is concentrated anywhere, and the matched and unmatched objects are interleaved.

V. VALIDATION

A. The validation contract

Table VI lists the requirements this release is held to. Each is machine-checkable against the published files; the checker and its raw output are released with the catalogue. Requirements cover provenance and hash binding (V1–V3b), the selection rule (V4, V7), the sky-fraction trend of Sec. IIIB (V9), internal consistency of the released tables (V5, V6, V8), three tests of what the score measures (V10–V12), and three defects the release must state rather than hide (DEFECT-1 to DEFECT-3).

Eleven requirements pass outright (V1–V10, V3b). V11 and DEFECT-3 are REPORTED quantities rather than thresholds. V12 returns STRUCTURED-BUT-WEAK, and we take the weaker reading throughout (Sec. VIC). DEFECT-1 and DEFECT-2 FAIL by design: they exist to force the release to carry its own known defects (Sec. VE). V1 is a partial pass: it verifies TARGETID > 0, the one of its three provenance-gate conditions that survives in the published file, because OBJTYPE and COADD_FIBERSTATUS are not carried in the released tables (Sec. VE records this as an open schema gap).

B. The score is not an exposure-quality proxy

The first thing to rule out is that the score measures how badly a spectrum was measured rather than what it contains. Within the science-target sample the anomaly score is essentially uncorrelated with every available measure of exposure quality: Spearman $\rho_s = +0.058$ against median coadd signal-to-noise, -0.076 against coadd exposure time, -0.093 and -0.096 against TSNR2_{LRG} and TSNR2_{QSO}, and -0.016 against r -band flux (Fig. 5). At $n = 1,244$, four of these reach $p < 0.05$; the three strongest are negative (longer exposures, higher TSNR2, score marginally *lower*), while the coadd signal-to-noise coefficient is positive and only marginal ($\rho_s = +0.058$, $p = 0.039$). All five are negligible in size.

The claim this supports is narrow and negative: the $S > 3$ tail is not explained by faintness or short exposure. It is not evidence that the tail is astrophysically

interesting. V10 should be read as closing one explanation, not as opening another.

C. The score is driven by the blue arm

DESI spectrographs record each fibre in three cameras, and the released catalogue carries the reconstruction residual of each separately. Decomposing by camera gives $\rho_s(S, r_B) = +0.527$ ($p \sim 10^{-89}$), against -0.037 ($p = 0.19$) for r_R and -0.142 ($p \sim 5 \times 10^{-7}$) for r_Z . The median residuals are correspondingly lopsided: $r_B = 2.53$ versus $r_R = 1.03$ and $r_Z = 0.94$. Because S is an exact affine function of `mean_mse` ($\rho_s(S, \text{mean_mse}) = 1.000$), $\rho_s(S, r_B)$ is a correlation between the score and one of its own components, not an independent measurement; a positive coefficient for whichever camera dominates the loss follows arithmetically. The right statement is a variance decomposition: regressing `mean_mse` on (r_B, r_R, r_Z) gives $R^2 = 0.777$; dropping r_B collapses it to $R^2 = 0.006$, while dropping r_R or r_Z leaves $R^2 = 0.609$ or 0.201 . This is distinct from what r_B explains in isolation: the univariate regression of `mean_mse` on r_B alone gives $R^2 = 0.198$ (against 0.001 for r_R and 0.004 for r_Z). The two readings are not interchangeable — one is a drop-one variance decomposition, the other a marginal fit — and both are stated explicitly wherever either is quoted, including in the abstract. The released `worst_band` column makes the same point directly: the b camera is the worst-fit camera for 1,180 of the 1,244 released objects (94.9%; 60 for z , 4 for r), and the released `peak_residual_wavelength` column concentrates hard against the DESI blue cutoff (median 4057 Å, minimum 3632 Å, 89.5% of objects below 5930 Å). Figure 6 shows the three per-camera relations. The b -camera panel is a near-monotonic locus; the other two are scatter.

Whatever the model responds to lives in the b camera. Two readings survive this measurement: blue-end sky-subtraction and calibration residuals, which are a known difficulty at the blue limit of fibre spectrographs, or genuine spectral behaviour at rest-frame wavelengths that land in the b camera for this redshift distribution. This is the single most consequential open question about the catalogue, because the first reading would make much of the list instrumental — and unlike the released columns used above, it is not fully undecided by the data already in hand. Two within-catalogue diagnostics both lean instrumental without closing the question. First, the b -camera residual has no detectable dependence on b -camera signal-to-noise ($\rho_s(r_B, \text{snr}_b) = -0.008$, $p = 0.78$; contrast $\rho_s(r_R, \text{snr}_r) = -0.078$, $p = 0.006$ and $\rho_s(r_Z, \text{snr}_z) = -0.171$, $p \sim 10^{-9}$), which disfavors the simplest reading that the b arm reconstructs worst because it is noisiest. Second, for the 502 objects with a secure pipeline redshift (`ZWARN` = 0), `peak_residual_wavelength` is more tightly concentrated in the *observed* frame (median 4211 Å, interquartile range 39% of the median) than in

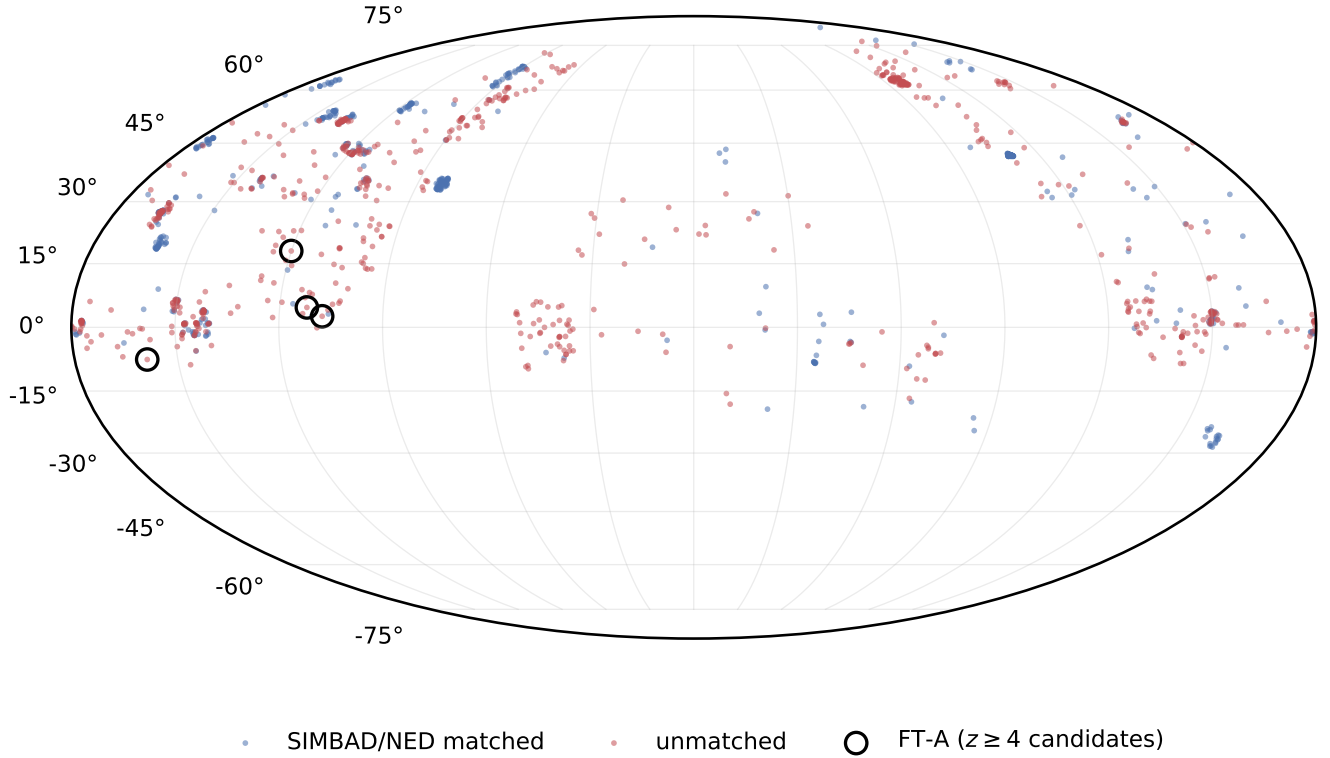


FIG. 4. Sky distribution of the released catalogue in equatorial coordinates, split by SIMBAD/NED counterpart status, with the four FT-A high-redshift candidates circled. The pattern traces DR1 tiling, not any property of the score.

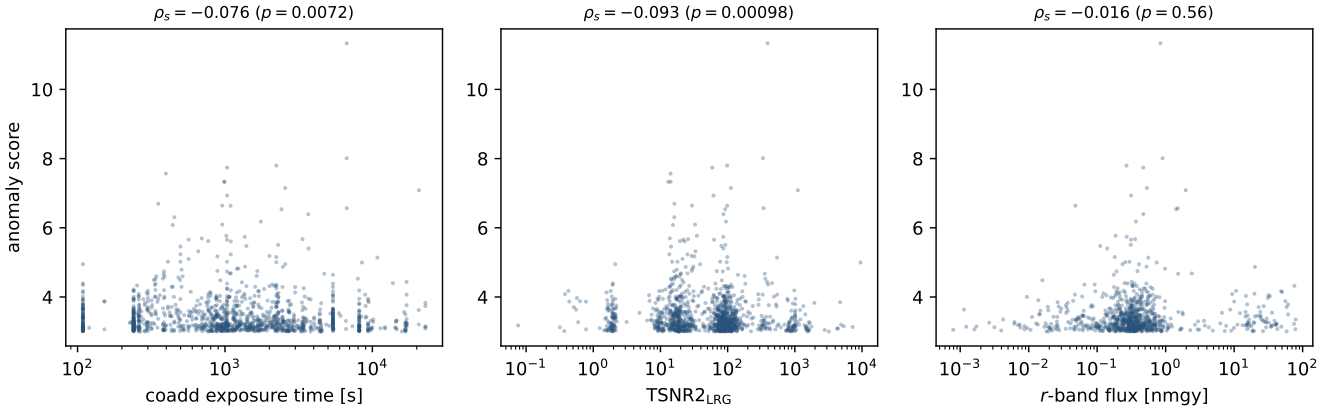


FIG. 5. Anomaly score against exposure time, $\text{TSNR2}_{\text{LRG}}$, and r -band flux for the 1,244 released objects, with Spearman coefficients.

the rest frame after de-redshifting (median 2098 Å, interquartile range 68% of the median): a genuine rest-frame spectral feature should pile up at a fixed rest wavelength, and a fixed-wavelength calibration artefact should pile up at a fixed observed wavelength, and the data show the latter pattern. These two diagnostics both favour the instrumental reading over the astrophysical one, but neither is a closed test: they rule out one specific noise-driven mechanism and one specific astro-

physical mechanism, not every member of either class. It is also cheap to test further on public data, and we name the airmass/lunar-phase/sky-subtraction-residual version of that test as OT-3 in Sec. VII C. The diagnostics in this section are computed by the committed script `blue_arm_diagnostics_2026-09-19.py` (Sec. IX), run on the released tables alone, with no new scan and no network access.

TABLE VI. Validation contract. Outcomes are re-run against the released artifacts; REPORTED marks a quantity the release must state rather than a threshold it must clear.

ID	Requirement	Outcome
V1	gates/check_sample_provenance.py exits 0 on the released sample tables (TARGETID > 0; OBJTYPE/FIBERSTATUS where carried)	PASS
V2	every released artifact's SHA-256 equals the value in the landing receipt (PHASE3_V2_LANDING_2026-09-03.md)	PASS
V3	the manifest chain closes end to end: sample $\rightarrow$ enrichment $\rightarrow$ cross-match $\rightarrow$ taxonomy, each stage binding the exact SHA-256 of its input and output, under one run contract, with the model / inference-code / zcatalog hashes recorded	PASS
V3b	the enrichment stage's own MSE reproduction cross-check passed for every released row	PASS
V4	replaying the pre-declared threshold rule (largest grid point with science-only count ≥ 300 , step up if $> 1,500$) on the science-target counts selects the released threshold	PASS
V5	the cross-match output partitions the sample exactly: matched + unmatched = sample, disjoint, no duplicate TARGETIDs, and the enriched table covers the same TARGETIDs	PASS
V6	the taxonomy covers exactly the SIMBAD/NED-unmatched subset; 23 clusters roll up into 9 families; per-family counts sum to the unmatched count and match the per-object labels	PASS
V7	every released object exceeds the stated score threshold ($S > 3$)	PASS
V8	the AllWISE table covers every released object and its match count equals the released figure (74)	PASS
V9	the released sky-fraction-by-score validation supports the stated claim that every score bin above 3 is $> 99.5\%$ sky-or-nonscience before the provenance gate	PASS
V10	within the science-target sample the anomaly score is not a restatement of exposure quality or source brightness ($ \text{Spearman } \rho < 0.2$ against S/N, exposure time, TSNR2 and r-band flux)	PASS
V11	the per-band residual decomposition identifies which camera arm drives the score (reported, not gated)	REPORTED
V12	the released taxonomy's family partition is tested for structure in the 128-dimensional latent space it was NOT built from (silhouette vs a label-permutation null, 50 draws)	STRUCTURED-BUT-WEAK
DEFECT-1	the enriched table's zcatalog columns (z, zwarn, spectype, subtype) are populated	FAIL
DEFECT-2	every derived residual-diagnostic column in the enriched table carries values	FAIL
DEFECT-3	the fraction of released objects carrying a DESI-reliable redshift (ZWARN == 0) is stated with the catalogue	REPORTED

D. Known-object recovery

A candidate list can be tested against classes of genuinely unusual objects that are already published: if the score is finding unusual astrophysics, it should preferentially recover them. We cross-matched the released sample at $1.5''$ against eleven candidate reference classes, of which five had usable positions (four lacked identifiable coordinate columns, one had no known catalogue identifier, and one timed out). The bar was declared in advance: at least one class recovered at $> 10\times$ enrichment with ≥ 5 matches.

Table VII gives the outcome, with enrichment computed against the 21,793,550-object science-target parent (Sec. II), the same denominator used for every other selected fraction in this paper. One broad-absorption-line quasar [26] matches, at $3.3\times$ the base rate; a single match on a $N_{\text{ref}} = 5,285$ reference class, so this enrichment carries a wide interval of its own and supports no standalone claim. Roma-BZCAT blazars [27], cataclysmic variables [28], Lyman- α emitters [29] and superluminous-supernova hosts [30] return nothing. *No reference class is recovered at the pre-declared bar*, and there is no closed-loop candidate class.

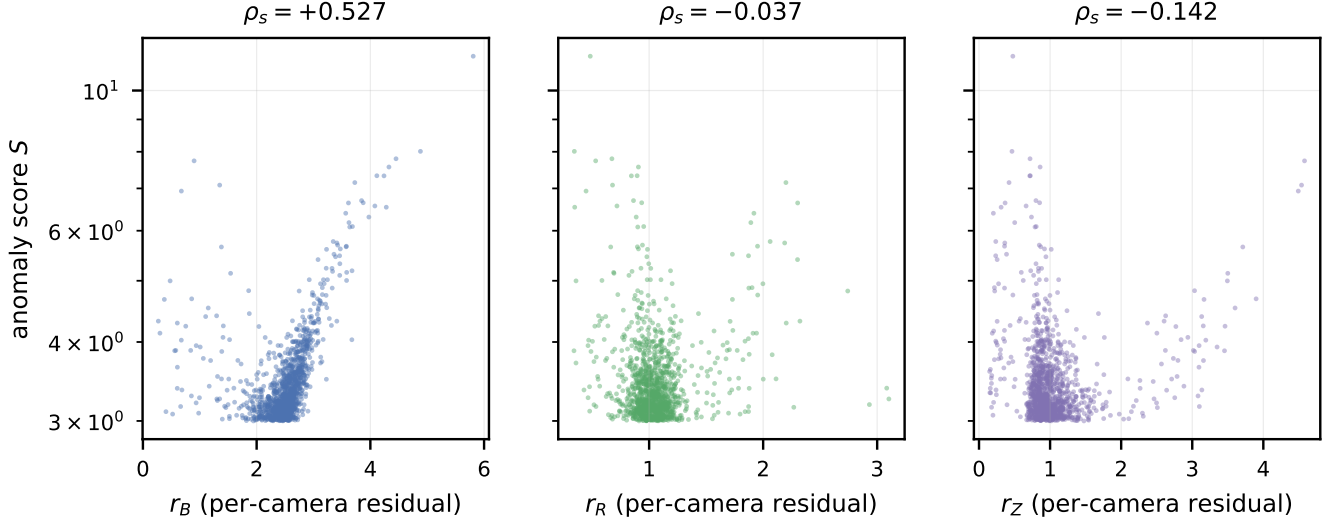


FIG. 6. Anomaly score against the per-camera reconstruction residual for the 1,244 released objects. The score tracks the b -camera residual; its dependence on r_R is consistent with zero ($\rho_s = -0.037$, $p = 0.19$) and its dependence on r_Z is small but significant ($\rho_s = -0.142$, $p \approx 5 \times 10^{-7}$).

This should sharpen rather than soften that conclusion: the reference footprints are small compared with the parent survey, so the test has little power for rare classes. The 95% upper limit on the recovery of superluminous-supernova hosts is 12.5%, which excludes almost nothing. The honest summary is that the benchmark rules out a strong recovery signal and is consistent with anything weaker.

This is the result that makes the present work a data release with a validated selection rather than a discovery paper, and we state it as such.

E. Known defects

Two defects in the released tables are recorded openly and must be repaired before public deposit. The enriched table's `z`, `ZWARN`, `SPECTYPE` and `SUBTYPE` columns are unpopulated, so a reader taking that file alone would read 1,244 null redshifts; the authoritative values are present in the sample table and are re-joined for every number in this paper, and the repair table is released alongside it. The `residual_kurtosis` column is empty; it is used by no result here and should be either populated or dropped from the published schema. Two further schema gaps are recorded in Sec. IV: the absent flux inverse variances, and the absent `OBJTYPE` and `COADD_FIBERSTATUS` columns, whose absence means the provenance gate cannot re-verify its own first two conditions from the published file alone. The deposit must carry all four.

VI. DESCRIPTIVE TAXONOMY

A. Construction

The 675 unmatched objects are grouped by PCA $\rightarrow$ UMAP [18] $\rightarrow$ HDBSCAN [19] on four standardised features (zero mean, unit variance): the sky position embedded on the unit sphere, $(\cos \alpha \cos \delta, \sin \alpha \cos \delta, \sin \delta)$, together with the anomaly score S , with k -nearest-neighbour noise reassignment, giving 23 clusters that roll up by shared (score-tier, dominant survey/programme) identity into 9 families (Table VIII, Fig. 7); two families can share the same printed (Tier, Survey/prog.) summary in Table VIII while remaining distinct, since the roll-up uses each cluster's own descriptor before aggregation, not the family-level summary shown there. $n_{\text{PCA}} = 10$ components are requested but clipped to the 4 available features, so PCA performs a rotation only, not a dimensionality reduction. Family sizes sum to 675 exactly, and the per-family counts agree with the per-object labels (check V6). Every hyperparameter and the random seed (`pca_components=10`, `umap_neighbors=15`, `umap_min_dist=0.05`, `min_cluster_size=15`, `min_samples=3`, `random_state=42`) are stated here and reused verbatim from the published run's own configuration file (`flagship_taxonomy_v2_manifest.json`), so this result is reproducible from the released files and this manuscript alone.

An earlier construction of this taxonomy fed raw, unstandardised (S, α, δ) into the same pipeline. Right ascension is periodic at 360° , so naive Euclidean distance on raw degrees splits objects near $\alpha \approx 0$ into maximally distant points; that earlier construction's largest family

TABLE VII. Known-object recovery benchmark, $1.5''$ positional match. N_{ref} counts reference objects inside the DR1 footprint; recovery is the matched fraction of those, with a 95% Wilson interval; enrichment is recovery over the catalogue’s base rate against the 21,793,550 science-target parent (Sec. II), not the 27,547,223 all-TARGETID parent.

Reference class	N_{ref}	N_{match}	Recovery (95% CI)	Enrichment
Broad absorption line (BAL) quasars	5,285	1	0.019% [0.003, 0.107]	$3.3\times$
Roma-BZCAT blazars (5th edition)	2,060	0	0.000% [0.000, 0.186]	$0.0\times$
Cataclysmic variables / white-dwarf binaries	580	0	0.000% [0.000, 0.658]	$0.0\times$
Lyman- α emitters (LAEs)	84	0	0.000% [0.000, 4.373]	$0.0\times$
Superluminous supernova (SLSN) host galaxies	27	0	0.000% [0.000, 12.456]	$0.0\times$

had a naive RA span of 350.4° — the signature of exactly this defect, not a genuinely sky-spanning family — and its hyperparameters and seed, while present in its own configuration file, were never stated in that construction’s own manuscript prose. A corrected re-clustering, run on the identical 675-object input with every hyperparameter and the seed reused verbatim from that earlier run (isolating the sky-embedding fix as the only changed variable; confirmed deterministic under a bit-identical rerun), and compared against a (survey, programme)-only baseline, produces the 23 clusters/9 families reported here: Adjusted Rand Index 0.60 and Adjusted Mutual Information 0.63 against the earlier, hash-bound partition (`flagship_taxonomy_v2.json`, Table VI) over the same objects — a moderate restructuring, not a dissolution: most of the score-tier $\times$ survey/programme structure survives, but no earlier family is reproduced unchanged and specific cluster boundaries are provisional. Against the (survey, programme)-only baseline the corrected families score ARI 0.26/AMI 0.26 — weaker agreement, confirming the score axis contributes real partition information beyond a bare survey/ programme relabelling, alongside the survey/programme structure quantified directly below. Critically, fixing the wrap does not make the largest family sky-compact: its true, wrap-aware RA span is still 331.6° (Table IX), so the not-a-compact-sky-region conclusion below is not an artifact of the earlier defect and is, if anything, reinforced by a correctly-measured span. Full method, hyperparameters, and the per-family comparison table: [reclustering_2026-09-22](#) (output SHA-256 `9cc3666f181c...`, dated 2026-09-22 — a separate, later provenance record from the sealed selection chain of Table VI, which binds only the 2026-09-03 landing). None of this affects the null result of Sec. VIC, which already reports the weaker STRUCTURED-BUT-WEAK reading throughout this paper, recomputed here against the corrected family labels.

B. What the families are

Because the clustering features are the score and the sky position, the families are in practice a score-tier $\times$ survey/programme stratification, of which the survey/programme axis alone already accounts for the

weaker ARI/AMI 0.26/0.26 reported above. Table IX makes the stratification explicit: 71.9% of objects lie in their family’s dominant survey/programme cell, and no family is a compact sky region: the smallest right-ascension span is 15.4° (family 7, whose own declination span is 18.6°), the smallest declination span is 12.3° (family 8, whose own right-ascension span is 30.8°), and the largest family spans 331.6° in right ascension once the wrap is corrected (Sec. VI) — none of the nine families is compact in both coordinates simultaneously.

C. The latent-space test

A stratification built from the score and the sky is not, on its face, a claim about spectra. The score S is itself an affine function of the 128-dimensional latent reconstruction error, so the clustering did see a scalar summary of that space; what it did not see is the space’s 128-dimensional *geometry*. We tested the partition against that geometry directly: its silhouette over the 128 latent dimensions is -0.0183 , against a 10,000-draw label-permutation null of -0.0310 ± 0.0063 whose largest draw is -0.0144 (check V12, Fig. 8), recomputed against the corrected family labels of Sec. VI. The observed value exceeds every one of the 10,000 permutation draws, so the empirical $p = 0.0041$: the partition is distinguishable from a random relabelling of the same score-derived quantity, though this test cannot by itself separate structure attributable to S from structure in the sky coordinates.

But the absolute silhouette is *negative*. A negative silhouette means the typical object is, on average, closer to some other family’s members than to its own. The families are therefore not separated clusters in latent space, and the correct reading is the weaker one: they are descriptive strata of the candidate list that carry a small amount of latent-space information, not physical classes. They are labelled with descriptors only, and no family is given a physical class name. Re-clustering on the latents themselves is the obvious next version (OT-2).

TABLE VIII. Per-family evidence over the 675 SIMBAD/NED-unmatched objects. Tier is a quantile cut of the 675-object score distribution (low/elevated/ high/extreme at the 50th/80th/95th percentile), not a released column. Survey/prog. is the family’s empirical dominant (survey, programme) pair, verified directly against member objects (not the per-object descriptor string, which can disagree with it; Sec. VI). N_{cl} is the number of source clusters, r_B the median b -camera residual, N_{Z0} the number with $Z\text{WARN} = 0$, $\bar{z}_{Z0}$ their median redshift restricted to $Z\text{WARN} = 0$, $N_{z>2}$ the number with $Z\text{WARN} = 0$ and $z > 2$, N_{QSO} the number with pipeline SPECTYPE = QSO over *all* family members regardless of $Z\text{WARN}$ (a classification, not a redshift measurement), and N_W the number with an AllWISE counterpart.

Family	Tier	N	N_{cl}	Survey/prog.	Score med. (5–95%)	r_B	N_{Z0}	$\bar{z}_{Z0}$	$N_{z>2}$	N_{QSO}	N_W
0	low	308	11	main/dark	3.14 (3.02–3.44)	2.42	160	1.15	20	25	0
1	elevated	81	3	main/dark	3.38 (3.14–3.97)	2.53	35	1.20	3	4	1
2	high	72	2	sv3/bright	3.81 (3.39–4.87)	2.72	19	1.35	1	3	2
3	low	56	1	sv1/other	3.17 (3.02–3.48)	2.38	17	0.92	0	2	2
4	low	54	2	sv3/bright	3.15 (3.01–3.82)	2.48	9	1.35	1	1	1
5	high	36	1	main/dark	3.62 (3.44–4.31)	2.81	27	1.09	0	1	0
6	extreme	31	1	sv3/bright	4.68 (3.92–6.52)	3.12	9	1.19	0	0	0
7	elevated	21	1	sv3/bright	3.48 (3.36–3.75)	2.53	5	1.50	2	3	0
8	low	16	1	sv1/other	3.12 (3.01–3.28)	2.31	8	1.50	2	3	0

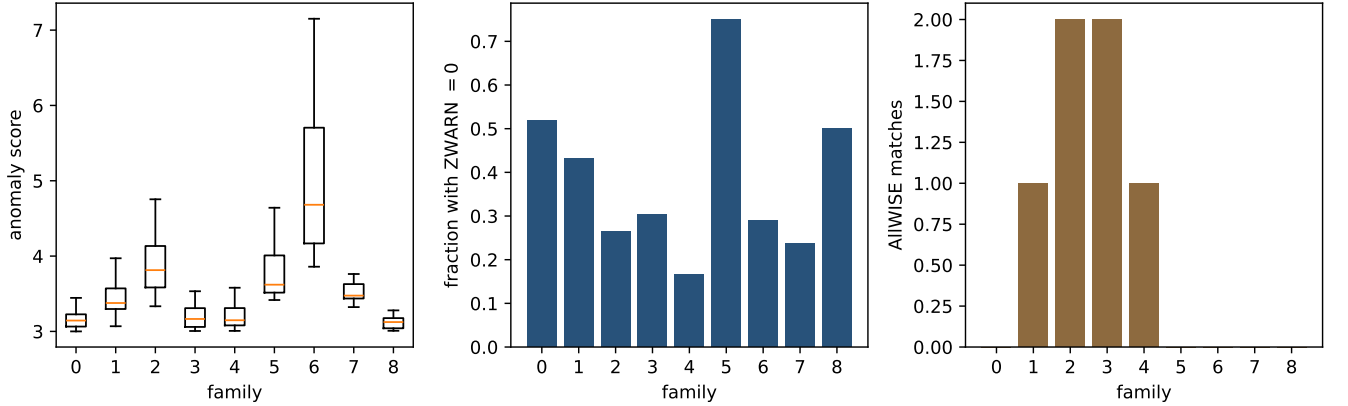


FIG. 7. Per-family anomaly-score distribution, reliable-redshift fraction, and AllWISE match count.

TABLE IX. Survey purity and sky extent of the nine families. Sky position enters the clustering only as a proxy for survey and programme footprint. R.A. spans are wrap-aware (Sec. VI).

Family	N	Dom. share	R.A. span	Dec. span
0	308	80.5%	331.6°	94.7°
1	81	87.7%	204.1°	81.2°
2	72	51.4%	147.1°	60.5°
3	56	37.5%	48.9°	20.5°
4	54	79.6%	42.9°	42.4°
5	36	75.0%	51.7°	23.3°
6	31	51.6%	69.9°	31.4°
7	21	57.1%	15.4°	18.6°
8	16	62.5%	30.8°	12.3°

D. Two families worth starting with

Two families stand out for follow-up purposes. Family 6 (31 objects, a single cluster) is the extreme-score family, with median $S = 4.68$, a 95th percentile of 6.52, the largest median blue residual of any family ($r_B = 3.12$), and the highest-scoring unmatched object in the catalogue ($S = 7.15$). Family 5 (36 objects) has the highest reliable-redshift fraction (27/36, 75%) at a high median score of 3.62 and no object above $z = 2$: well-measured, low-redshift, high-score, and therefore the family in which an instrumental or template explanation is most directly testable. Family 2, the same score tier’s worst redshift reliability (19/72, 26%), is not its mirror in every other respect: unlike the corresponding comparison under the earlier construction, it carries 3 pipeline SPECTYPE = QSO classifications (not necessarily at $Z\text{WARN} = 0$) and 2 AllWISE matches, so any follow-up there should start from those existing identi-

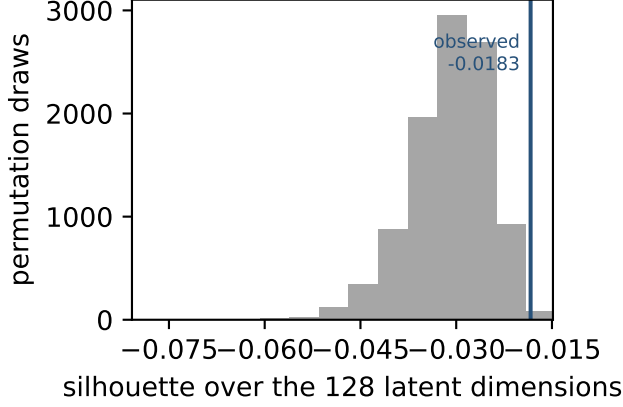


FIG. 8. Silhouette of the nine-family partition over the 128 latent dimensions, against its 10,000-draw label-permutation null. The observed value beats every draw but is itself negative: structured, but not separated.

cations rather than from re-reduction alone.

The families also inherit the limits of the cross-match. AllWISE coverage of the unmatched subset is 6 of 675, so mid-infrared colours cannot discriminate between families at all; the 74 AllWISE matches in the full sample sit overwhelmingly in the matched half, which is expected, since catalogued objects are catalogued because they are detectable elsewhere.

VII. FOLLOW-UP SET AND OPEN TESTS

A. Named targets

Table X names 30 distinct objects across 33 tier rows in five rules-based tiers, all drawn from the unmatched subset. The rules were fixed from literature criteria rather than tuned to yield a target count, and are replayable from the released files.

- **FT-A** (4 targets) — $\text{ZWARN} = 0$ and $z \geq 4$: four anomaly-selected DESI-pipeline $z \geq 4$ quasar candidates with no catalogued counterpart, two of them at $z > 6$. These are pipeline redshifts, not confirmations. Section VII B runs the cheapest discriminating test on them here.
- **FT-B** (1 target) — AllWISE [8] $W1 - W2 \geq 0.8$, the standard mid-infrared AGN criterion [20], measured on the WISE mission [21]. The object is TARGETID 103610381762578 at $W1 - W2 = 0.822$, in family 3. Next tests are public archives: NEOWISE-R light curve [22], ZTF optical light curve [25], archival X-ray coverage.
- **FT-C** (5 targets) — $\text{ZWARN} = 0$, $\Delta\chi^2 > 25$ and S in the top 5%: spectra that are simultaneously well measured and hardest to reconstruct, where a genuine

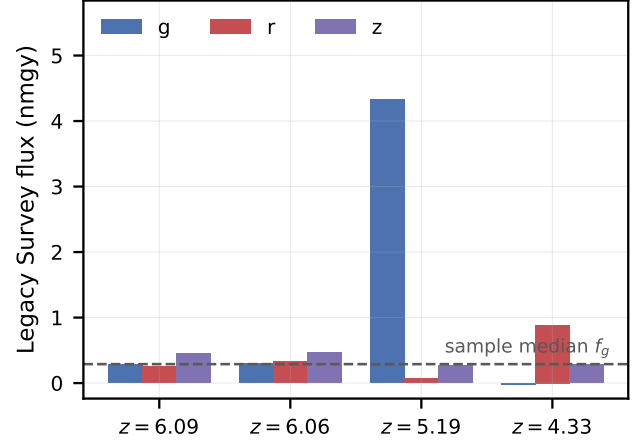


FIG. 9. Legacy Survey *grz* fluxes of the four FT-A candidates. A Lyman break at the pipeline redshift requires a *g*-band dropout *and* a detection in the reddest band fully blueward of the break; the $z = 5.19$ candidate's *g* flux instead sits at the sample's 94.7th percentile.

spectral anomaly is most likely. Scores run from 4.64 to 7.15 and three of the five are in family 6.

- **FT-D** (0 targets) — unmatched point sources with a Gaia DR3 counterpart [24]. The rule returns nothing, and that is a result: none of the 675 unmatched objects has a Gaia counterpart at all, so Galactic follow-up of this catalogue is not cheap. We retain the rule so a deeper generation can re-run it.
- **FT-E** (23 targets) — the highest-score member of each cluster, so that a follow-up programme tests the taxonomy rather than one corner of it.

B. A Lyman-break check on the FT-A candidates

The cheapest test in the whole follow-up set costs no observing time. A quasar at redshift z places Lyman- α at $1216(1+z)$ Å, and the intergalactic medium absorbs essentially everything blueward of it, so any broadband filter lying entirely below the break must go dark. For the four FT-A candidates the break falls at 6484, 7530, 8587 and 8616 Å, so the Legacy Survey *g* band (≈ 4000 – 5500 Å) is fully blueward of the break for all four, and the *r* band (≈ 5500 – 7000 Å) for the three at $z > 4.5$. The released photometry already contains the answer. Table XI and Fig. 9 give it.

Two outcomes, stated at their evidential strength. A correct break test needs both halves for every candidate it applies to: a dropout in every band fully blueward of the break, and a surviving detection in the reddest band that straddles or sits just redward of it. We apply that test uniformly below, which changes one of this section's conclusions relative to the released *g*-only check.

TABLE X. Named follow-up targets. S is the anomaly score. Redshifts are DESI pipeline values and are printed only where $\text{ZWARN} = 0$; a dash marks an object whose fit carries $\text{ZWARN} = 4$ (`SMALL_DELTA_CHI2`), for which the pipeline redshift is not a measurement. The FT-E tier spans the taxonomy by construction and therefore includes such objects.

Tier	TARGETID	R.A.	Dec.	S	z	ZWARN	SPECTYPE
FT-A	39627903570285763	247.89529	4.66991	3.496	6.0878	0	QSO
FT-A	39627607293039577	201.10137	-7.55334	3.232	6.0638	0	QSO
FT-A	39628216868014338	239.92839	18.01577	3.061	5.1938	0	QSO
FT-A	39627849358909321	252.47160	2.60073	3.033	4.3338	0	QSO
FT-B	103610381762578	223.91434	50.21732	3.105	—	4	GALAXY
FT-C	39627812453222421	212.59529	0.92095	7.151	1.0798	0	GALAXY
FT-C	39627812461610773	213.07783	0.96584	6.394	1.1868	0	GALAXY
FT-C	39627877817258319	149.92219	3.77396	6.180	1.1989	0	GALAXY
FT-C	39633001079899014	217.98717	36.00150	4.747	1.3533	0	GALAXY
FT-C	39633407088529181	97.49930	61.65664	4.642	1.3098	0	GALAXY
FT-E	39627812453222421	212.59529	0.92095	7.151	1.0798	0	GALAXY
FT-E	1070924837158914	252.78766	35.77081	5.320	—	4	GALAXY
FT-E	39633407088529181	97.49930	61.65664	4.642	1.3098	0	GALAXY
FT-E	1070131593609217	149.69873	0.83556	4.565	—	4	GALAXY
FT-E	39636182677589446	64.05184	-16.74091	4.321	0.0063	0	GALAXY
FT-E	39627748041295714	333.31949	-1.73961	4.252	1.4203	0	GALAXY
FT-E	39633369285264389	275.33969	58.53970	4.115	—	4	GALAXY
FT-E	1070759501889569	194.40490	27.75104	3.963	—	4	GALAXY
FT-E	1070259272417281	210.20609	6.12193	3.762	—	4	GALAXY
FT-E	39627799471853674	158.98598	0.57838	3.617	—	4	GALAXY
FT-E	39628361990931280	150.95167	24.43648	3.593	0.9264	0	GALAXY
FT-E	1070874476150795	217.24189	33.16795	3.533	—	4	GALAXY
FT-E	39627903570285763	247.89529	4.66991	3.496	6.0878	0	QSO
FT-E	39628507847853660	255.35261	31.36383	3.426	1.2814	0	GALAXY
FT-E	39633418778053415	143.59822	62.82238	3.416	1.3659	0	GALAXY
FT-E	39633432057219468	96.54153	63.88652	3.387	0.9304	0	GALAXY
FT-E	39628387353889221	17.03117	25.76763	3.363	1.3410	0	GALAXY
FT-E	39627817989705740	182.69435	1.34328	3.325	1.2954	0	GALAXY
FT-E	39633062929108448	182.23513	39.17878	3.279	0.2303	0	GALAXY
FT-E	39633426692703230	98.75150	63.52515	3.264	—	4	GALAXY
FT-E	1070090426515458	215.89907	-1.05994	3.247	—	4	GALAXY
FT-E	39633184245159076	270.34970	45.93852	3.194	1.4010	0	GALAXY
FT-E	39627808204394328	319.33259	0.65412	3.189	—	4	GALAXY

TABLE XI. The four FT-A candidates. Fluxes are Legacy Survey values in nanomaggies as carried in the released catalogue; the sample-wide median f_g is 0.289 nmgy.

TARGETID	z	S	$\Delta\chi^2$	f_g	f_r	f_z
39627903570285763	6.0878	3.496	23.4	0.283	0.255	0.446
39627607293039577	6.0638	3.232	135.5	0.296	0.330	0.470
39628216868014338	5.1938	3.061	22.0	4.326	0.073	0.262
39627849358909321	4.3338	3.033	4205.1	-0.019	0.886	0.286

Refuted: TARGETID 39628216868014338 ($z = 5.194$) carries $f_g = 4.33$ nmgy (94.7th percentile of the whole

catalogue; 66 objects, including itself, carry a g flux at least this bright), with $f_g/f_r = 59$. Both g and r lie fully blueward of a $z = 5.19$ break, so a 4.33 nmgy g -band flux ($m = 22.5 - 2.5 \log_{10}(4.33) \approx 20.9$ AB mag, orders of magnitude above any plausible Legacy Survey noise floor) is inconsistent with the pipeline redshift on the physics alone, independent of any percentile ranking within this catalogue. Its $\Delta\chi^2$ of 22.0 is also the weakest of the four, and its morphology is **SER** rather than point-like. We classify it as *disfavoured by public imaging* — either the photometry is blended or the redshift solution is wrong — and retain it in the table so that the tier rule can be audited.

Undecidable from the release: The other three candidates cannot be settled by the released photometry, for two distinct reasons. **TARGETID** 39627849358909321 ($z = 4.334$; $\Delta\chi^2 = 4205$, by far the strongest redshift fit in the tier) carries $f_g = -0.019$ nmgy (consistent with zero, as the g -band half of the test requires) but also `morphtype` = empty and $f_{W1} = f_{W2} = \text{sersic} = \text{shape_r} = 0$ exactly — the exact signature of the 406-row photometric-join gap disclosed in Sec. IV, not a clean measurement. Its own r - and z -band fluxes ($f_r = 0.886$ nmgy, $f_z = 0.286$ nmgy, $f_r/f_z = 3.1$) are backwards for the redshift they would need to support: at $z = 4.334$ the r band is $\sim 66\%$ forest-absorbed while z is unabsorbed continuum on a red-rising spectrum, so the expected ordering is $f_r \lesssim f_z$, not $f_r \approx 3f_z$. We do not adopt either reading; the object needs the same forced photometry as the other two before it can be classified either way. The two $z \approx 6.07$ candidates carry $f_g = 0.283$ and 0.296 nmgy — 0.98 and 1.02 times the catalogue median g flux — values consistent with either a faint detection or a non-detection. Neither has a photometric-join flag, but both also carry $f_r = 0.255$ and 0.330 nmgy, comparable to or larger than their own g fluxes in a band the break requires to be dark, and f_z/f_r ratios ($1.75, 1.42$) far shallower than a Gunn–Peterson trough at $z \approx 6$ would produce. Because the released schema omits flux inverse variances for all three objects, none of these values can be distinguished from noise, so none of the three can be confirmed or refuted here. Legacy Survey DR10 forced photometry with inverse variances, applied to all four bands for all three objects, settles this at zero observing cost and is the immediate next step; until then this tier’s verdict is *one refuted, three undecided, not one refuted, one supported, two undecided*.

We emphasise what this does and does not establish. One candidate has been removed by public data alone, and none has cleared the full break test on the photometry released here; this makes no object in this catalogue a discovery, and a **ZWARN** = 0 Redrock solution on a low-signal-to-noise spectrum fails routinely at high redshift. The remaining tests, in cost order, are: the DR10 forced-photometry query above; visual inspection of the public DR1 coadd for a Lyman- α break and forest; DESI DR2 repeat coadds where the tile is revisited; and only then confirmation spectroscopy, optical for the $z \approx 4$ candidate and near-infrared for the $z \approx 6$ pair.

C. Open tests

OT-1 Selection function: No injection-recovery experiment has been run for this model on this substrate. The catalogue therefore has no measured completeness or purity, and no population-level inference may be drawn from it. This is the gate any rate, abundance, or cosmological use must clear first, and it requires injecting known spectral classes into DR1 coadds at controlled

signal-to-noise and re-scoring with the hash-bound model — GPU-scale work, not local, and genuinely not run for this release. The experiment is fully specified (inject each of the five reference classes’ template spectra from Table VII at a grid of signal-to-noise ratios into real DR1 sky/noise realizations, re-score with `best_model.47k.pt` under the same run contract, and report per-class recovery vs. injected signal-to-noise) but not executed; we state the specification rather than a placeholder so a reader can judge the remaining work.

OT-2 Latent-space taxonomy: Re-clustering on the 128 latent dimensions rather than on (S, α, δ) , following the negative silhouette of Sec. VIC. Cheap; the released catalogue already carries the latents.

OT-3 Blue-arm diagnosis: Compare the b -camera residual against airmass, lunar phase, and blue-end sky-subtraction residuals in the DR1 exposure tables, to separate calibration from astrophysics. This decides how much of the catalogue is astrophysical and is the highest-value open test after OT-1.

OT-4 Wider recovery benchmark: Six of eleven candidate reference classes lacked usable positions and were not tested; recovering their coordinates would materially strengthen or weaken Sec. VD.

OT-5 Deeper scan: A self-supervised model trained on the full science-target spectrum population, rather than the archived autoencoder (whose training-set size and composition this release does not document, Sec. III), is the realistic route to a confirmed anomalous class.

VIII. WHAT THIS CATALOGUE BEARS ON

This catalogue was built inside a programme whose interest in DESI spectra is an early-universe one: assembling anomalies in existing public data that strain single-field inflation, and attaching an explicit discriminator to each. It is worth being precise about how little of that this release supports, because the temptation to over-connect a candidate list to a cosmological question is exactly what a provenance-first release exists to resist.

Three statements are defensible. First, the catalogue is *infrastructure* for such a programme rather than evidence within it: it supplies a reproducible, hash-bound path from a public survey to a candidate list, and a validation contract a third party can re-run. Second, the only channel by which this particular catalogue could feed an early-universe abundance argument is its high-redshift end — the 36 reliable-redshift objects at $z \geq 4$ and 16 at $z \geq 6$, of which the four unmatched FT-A candidates are the interesting subset. Third, that channel is closed at present, and for two independent reasons: the candidates are unconfirmed, and Sec. VII B has already removed one of four on public data alone; and without OT-1 the catalogue has no selection function, so it ad-

mits no abundance, rate, or number-density statement of any kind.

We also record a null from the same programme that bounds what a catalogue like this could ever contribute. Local-shape primordial non-Gaussianity, parameterised by the amplitude f_{NL} of the bispectrum $B_{\Phi} = 2f_{\text{NL}}[P(k_1)P(k_2) + 2 \text{ perms}]$, shifts the abundance of rare, high-redshift haloes through the tail of the mass function. For the amplitude this programme’s matter-dominated-contraction bounce predicts ($f_{\text{NL}} = -35/16 = -2.1875$), applying the LoVerde, Miller, Shandera & Verde (2008) [2] non-Gaussian correction to a Planck-2018 [4] Λ CDM mass function (Eisenstein & Hu 1998 [3] transfer function) shifts the $z \approx 10$ –14, $\log M_h/M_{\odot} \approx 11.5$ halo abundance by -6% to -16% relative to the Gaussian case — a suppression, not an enhancement, so it cannot explain the reported early-galaxy over-abundance [1, 5]. That shift is a factor of roughly 1.06 to 1.19, well inside the $\gtrsim 0.3$ dex ($\approx 2\times$) stellar-mass systematic floor of current high-redshift photometric censuses, not more than an order of magnitude below it; a factor-of-two abundance effect at this halo mass and redshift would require $|f_{\text{NL}}| \sim 30$, far outside current CMB and large-scale-structure constraints. The full computation is the committed, deterministic script `ledger6_png_highz_abundance.py` (no fitting, no randomness; SHA-256 `f1121e2b79c4...`), whose output is `ledger6_png_highz_abundance.json`. The high-redshift abundance is therefore not where that class of model is measurable, whatever a candidate catalogue contributes to it. Publishing that null beside the catalogue is the point: it is what keeps the release from being read as evidence it is not.

IX. DATA AVAILABILITY AND REPRODUCIBILITY

The catalogue, its manifests, the validation checker and its output, the per-family evidence tables, the follow-up set, and the figure and table generators are released in the project repository under `results_2026-08-07/-phase3_v2` and `flagship_assembly_2026-09-18` (both under `pipelines/p1_highz_tracers/`), with mirrors on Hugging Face and Backblaze B2 verified by checksum. The full reproducibility manifest — external data sources and links, APIs, scripts, compute venue, cost, and wall-clock — is [REPRODUCIBILITY MANIFEST.md](#). [PLACEHOLDER: Zenodo DOI for the public deposit]

Reproduction has two stages with very different costs.

Stage A — scan and characterisation. The scan, enrichment, cross-match, AllWISE query and taxonomy ran end to end on a single rented GPU pod at \$0.17/hr in 7h 42m 53s, for approximately **\$1.31** of compute. The dominant stages are not compute-bound: enrichment took 2h 43m and the SIMBAD/NED cross-match 3h 24m, the latter limited by public service rate limits, so it reproduces at similar wall-clock on any machine. Only the scoring pass requires a GPU. All DESI, SIM-

BAD, NED, VizieR and Legacy Survey inputs are public and need no account.

Stage B — every number in this paper. The assembly reads only the ten hashed artifacts, uses no network, and runs on a laptop CPU in under 60 s at essentially zero cost:

```
python3 assemble_flagship_evidence.py
python3 make_tables.py
python3 make_draft_figures.py
python3 blue_arm_diagnostics_2026-09-19.py
```

The taxonomy of Sec. VI (Tables VIII and IX, Figs. 7 and 8) is produced by a fourth, self-contained script that reruns the same joins and re-derives the family evidence, follow-up set, and latent-space test against the corrected clustering: `taxonomy_adoption_2026-09-22`, reading the clustering itself from `reclustering_2026-09-22` (CPU only, < 1 minute, \$0 cost, seed 42).

It emits the validation-contract outcome, the family evidence, the follow-up set, every table under `tables/`, every figure under `figures/`, and `outputs/draft_numbers.json`, which carries each derived quantity quoted in this text. Every number in this manuscript names its source artifact in a comment in the L^AT_EX source. The pipeline is fully deterministic; the V12 permutation null is seeded.

The archived autoencoder weights are a lab artifact, mirrored and hash-bound; retraining from scratch is not part of this reproduction path, while scoring is fully reproducible given the frozen weights. The selection function does not exist for this model–substrate pair (OT-1), and nothing in this paper depends on it.

X. WHAT THIS CATALOGUE IS NOT

- It is not a discovery catalogue. No object here is reported as a confirmed unusual source; selection-pipeline output alone never promotes an object to notable.
- It is not a statistically characterised sample. Without OT-1 there is no completeness, no purity, and no selection function, so no population inference is available.
- Its families are not physical classes (Sec. VIC).
- Its unmatched fraction is not a novelty fraction: it includes objects simply outside other surveys’ footprints or below their detection limits, and the population is faint enough that only six objects in the whole catalogue have a Gaia counterpart.
- Its anomaly score is a distance from one model’s learned reconstruction, not a physically motivated significance, and Sec. VC shows it is dominated by one spectrograph arm.
- Its high-redshift candidates are pipeline redshifts, not confirmations — and one of four has already failed a public-data check.

XI. CONCLUSIONS

We publish 1,244 DESI DR1 science-target spectra selected by reconstruction-error anomaly score — the top 5.7×10^{-5} of the 21.8 million science targets scanned — with a sealed provenance chain, a published validation contract of sixteen machine-checkable requirements, a descriptive taxonomy of the unmatched subset, and a named follow-up set.

Four results are worth carrying forward independently of the catalogue. A score cut without a provenance gate selects sky fibres, not objects, and the contamination worsens with score up to $S = 10$, above which the trend is unconstrained by a single sparse bin; this is a property of the public coadd substrate and applies to any reconstruction-error search run on it. Within the science-target sample the score is not a proxy for exposure quality or brightness. The score is almost entirely explained by the blue spectrograph arm’s own reconstruction residual (full three-camera model $R^2 = 0.78$, collapsing to 0.01 when the blue arm is dropped; the blue arm alone, univariately, explains $R^2 = 0.20$); that residual shows no dependence on blue-arm signal-to-noise and concentrates more tightly in the observed wavelength frame than the rest frame, evidence that leans instrumental without closing the question, and we name the decisive external test as OT-3. And the taxonomy’s families, tested in the model’s own latent space, are structured but not separated — descriptive strata, not classes.

The known-object recovery benchmark does not clear its pre-declared bar, but with reference samples as small as 27 objects it also has little power to have found a strong recovery signal (95% upper limit up to 12.5% for the rarest class); the one follow-up test cheap enough to run here refutes one of four high-redshift candidates on public imaging alone and cannot decide the other three. We therefore report the work as what it is: reproducible data infrastructure with a validated selection, a stratified candidate list, and a named list of the tests it has not yet passed.

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This research has made use of the SIMBAD database [17], operated at CDS, Strasbourg, France; the NASA/IPAC Extragalactic Database (NED), which is funded by the National Aeronautics and Space Administration and operated by the California Institute of Technology; and the VizieR catalogue access tool, CDS, Strasbourg, France, all queried through astroquery [37].

All computation for this manuscript’s assembled numbers (Appendix A) ran locally on the author’s own hardware at zero incremental cost. The upstream scan and characterisation stage (Sec. IX) used a single low-cost RunPod GPU rental (\$1.31 total); no proprietary or allocated compute time was used.

AI-assisted methodology.—AI coding and review tools, operated under the author’s direction, assisted with implementation review, documentation, manuscript drafting and revision, and adversarial cross-checking and adversarial review. Their outputs were not treated as evidence: all catalogue assertions reported here were accepted only after direct checking against the machine-readable artifacts, committed code, and local or public-source validation. The author set the study scope, made the scientific judgments, and takes full responsibility for the manuscript. No AI system is an author or is credited as one.

Appendix A: Software

The assembly uses NumPy [31], pandas [32], SciPy [33], scikit-learn [34], Matplotlib [35], Apache Arrow/PyArrow [36], and astroquery [37] for the SIMBAD, NED and VizieR queries. Exact versions are recorded in the reproducibility manifest.

The four citations not recorded in any committed benchmark artifact (Ref. [28]’s journal/volume/page, and the three software citations Refs. [34, 35, 37]) were verified against ADS for this manuscript (2026-09-19): Ritter & Kolb 2003, *Astron. Astrophys.* **404**, 301 (ADS bibcode 2003A&A...404..301R); Pedregosa *et al.* 2011, *J. Mach. Learn. Res.* **12**, 2825 (JMLR volume 12 official record); Hunter 2007, *Comput. Sci. Eng.* **9**, 90 (ADS

bibcode 2007CSE.....9...90H); and Ginsburg *et al.* 2019, *Astron. J.* **157**, 98 (ADS bibcode 2019AJ....157...98G, arXiv:1901.04520). All four entries below already carried the correct values; no bibliography text changed.

The remaining 26 entries below were separately ADS/journal-verified (2026-09-21): every journal, volume, and page matches the publisher’s or ADS’s record exactly, including the three least-common entries (Nicolaou *et al.* 2026, *Mon. Not. R. Astron. Soc.* **547**, page-ID stag010 — OUP’s continuous-publication identifier, not a typographical error; the DESI DR1 release paper, *Astron. J.* **171**, 285, 2026; and Guy *et al.* 2023, *Astron. J.* **165**, 144). No bibliography text changed as a result.

This appendix’s bibliography lists 37 entries in total. Of these, three carry no fixed journal record to check by design and are correctly cited as such: [9] (still “in preparation” per DESI’s own papers page), and [16, 36] (living documentation/URL citations, not journal articles). The remaining entries support the early-galaxy over-abundance comparison of Sec. VIII, among other results, and include [1] (*Nature* **616**, 266, 2023), [5] (*Nat. Astron.* **7**, 731, 2023), [2] (*J. Cosmol. Astropart. Phys.* **04**, 014, 2008), [3] (*Astrophys. J.* **496**, 605, 1998), and [4] (*Astron. Astrophys.* **641**, A6, 2020); every entry’s journal, volume, page, and year have been checked against ADS or the publisher’s own record.

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- [1] I. Labbé *et al.*, “A population of red candidate massive galaxies 600 Myr after the Big Bang,” *Nature* **616**, 266 (2023), arXiv:2207.12446.
 - [2] M. LoVerde, A. Miller, S. Shandera, and L. Verde, “Effects of scale-dependent non-Gaussianity on cosmological structures,” *J. Cosmol. Astropart. Phys.* **04**, 014 (2008), arXiv:0711.4126.
 - [3] D. J. Eisenstein and W. Hu, “Baryonic features in the matter transfer function,” *Astrophys. J.* **496**, 605 (1998), arXiv:astro-ph/9709112.
 - [4] Planck Collaboration, “Planck 2018 results. VI. Cosmological parameters,” *Astron. Astrophys.* **641**, A6 (2020), arXiv:1807.06209.
 - [5] M. Boylan-Kolchin, “Stress testing Λ CDM with high-redshift galaxy candidates,” *Nat. Astron.* **7**, 731 (2023), arXiv:2208.01611.
 - [6] This research made use of the NASA/IPAC Extragalactic Database (NED), funded by NASA and operated by the California Institute of Technology, <https://ned.ipac.caltech.edu/>.
 - [7] F. Ochsenbein, P. Bauer, and J. Marout, “The VizieR database of astronomical catalogues,” *Astron. Astrophys. Suppl. Ser.* **143**, 23 (2000).
 - [8] R. M. Cutri *et al.*, “VizieR Online Data Catalog: All-WISE Data Release,” *VizieR Online Data Catalog II/328* (2014).
 - [9] A. Bailey *et al.*, “Redrock: spectroscopic classification and redshift fitting for DESI,” in preparation; software <https://github.com/desihub/redrock>.
 - [10] D. Baron and D. Poznanski, “The weirdest SDSS galaxies: results from an outlier detection algorithm,” *Mon. Not. R. Astron. Soc.* **465**, 4530 (2017).
 - [11] Y. Liang *et al.*, “Outlier Detection in the DESI Bright Galaxy Survey,” *Astrophys. J. Lett.* **956**, L6 (2023), arXiv:2307.07664.
 - [12] C. Nicolaou *et al.*, “Identifying Anomalous DESI Galaxy Spectra with a Variational Autoencoder,” *Mon. Not. R. Astron. Soc.* **547**, stag010 (2026), arXiv:2506.17376.
 - [13] DESI Collaboration, “Data Release 1 of the Dark Energy Spectroscopic Instrument,” *Astron. J.* **171**, 285 (2026), arXiv:2503.14745.
 - [14] DESI Collaboration, “Overview of the Instrumentation for the Dark Energy Spectroscopic Instrument,” *Astron. J.* **164**, 207 (2022).
 - [15] J. Guy *et al.*, “The Spectroscopic Data Processing Pipeline for the Dark Energy Spectroscopic Instrument,” *Astron. J.* **165**, 144 (2023).
 - [16] DESI Data Systems, “Bit Masks in DESI: target masks and Redrock ZWARN,” DESI Data Model 25.3 (2025), <https://desidatamodel.readthedocs.io/en/25.3/bitmasks.html>.
 - [17] M. Wenger *et al.*, “The SIMBAD astronomical database,” *Astron. Astrophys. Suppl. Ser.* **143**, 9 (2000).
 - [18] L. McInnes, J. Healy, and J. Melville, “UMAP: Uniform Manifold Approximation and Projection for Dimension Reduction,” arXiv:1802.03426 (2018).
 - [19] R. J. G. B. Campello, D. Moulavi, and J. Sander, “Density-based clustering based on hierarchical density estimates,” in *PAKDD 2013*, Lecture Notes in Computer Science **7819**, 160 (2013).
 - [20] D. Stern *et al.*, “Mid-infrared selection of active galactic nuclei with the Wide-field Infrared Survey Explorer. I,” *Astrophys. J.* **753**, 30 (2012).
 - [21] E. L. Wright *et al.*, “The Wide-field Infrared Survey Explorer (WISE): mission description and initial on-orbit performance,” *Astron. J.* **140**, 1868 (2010).
 - [22] A. Mainzer *et al.*, “Initial performance of the NEOWISE reactivation mission,” *Astrophys. J.* **792**, 30 (2014).
 - [23] A. Dey *et al.*, “Overview of the DESI Legacy Imaging Surveys,” *Astron. J.* **157**, 168 (2019).
 - [24] Gaia Collaboration, “Gaia Data Release 3: summary of the content and survey properties,” *Astron. Astrophys.* **674**, A1 (2023).
 - [25] E. C. Bellm *et al.*, “The Zwicky Transient Facility: system overview, performance, and first results,” *Publ. Astron. Soc. Pac.* **131**, 018002 (2019).
 - [26] R. R. Gibson *et al.*, “A catalog of broad absorption line quasars in Sloan Digital Sky Survey Data Release 5,” *Astrophys. J.* **692**, 758 (2009).
 - [27] E. Massaro *et al.*, “The 5th edition of the Roma-BZCAT: a short presentation,” *Astrophys. Space Sci.* **357**, 75 (2015).
 - [28] H. Ritter and U. Kolb, “Catalogue of cataclysmic binaries, low-mass X-ray binaries and related objects (7th edition),” *Astron. Astrophys.* **404**, 301 (2003).
 - [29] M. Ouchi *et al.*, “Subaru/XMM-Newton Deep Survey. IV. Evolution of Ly α emitters from $z = 3.1$ to 5.7,” *Astrophys. J. Suppl. Ser.* **176**, 301 (2008).
 - [30] D. A. Perley *et al.*, “Host-galaxy properties of 32 low-redshift superluminous supernovae from the Palomar

- Transient Factory,” *Astrophys. J.* **830**, 13 (2016).
- [31] C. R. Harris *et al.*, “Array programming with NumPy,” *Nature* **585**, 357 (2020).
 - [32] W. McKinney, “Data structures for statistical computing in Python,” in *Proceedings of the 9th Python in Science Conference*, 56 (2010).
 - [33] P. Virtanen *et al.*, “SciPy 1.0: fundamental algorithms for scientific computing in Python,” *Nat. Methods* **17**, 261 (2020).
 - [34] F. Pedregosa *et al.*, “Scikit-learn: machine learning in Python,” *J. Mach. Learn. Res.* **12**, 2825 (2011).
 - [35] J. D. Hunter, “Matplotlib: a 2D graphics environment,” *Comput. Sci. Eng.* **9**, 90 (2007).
 - [36] Apache Software Foundation, “Apache Arrow and PyArrow,” <https://arrow.apache.org/docs/>.
 - [37] A. Ginsburg *et al.*, “astroquery: an astronomical web-querying package in Python,” *Astron. J.* **157**, 98 (2019).